



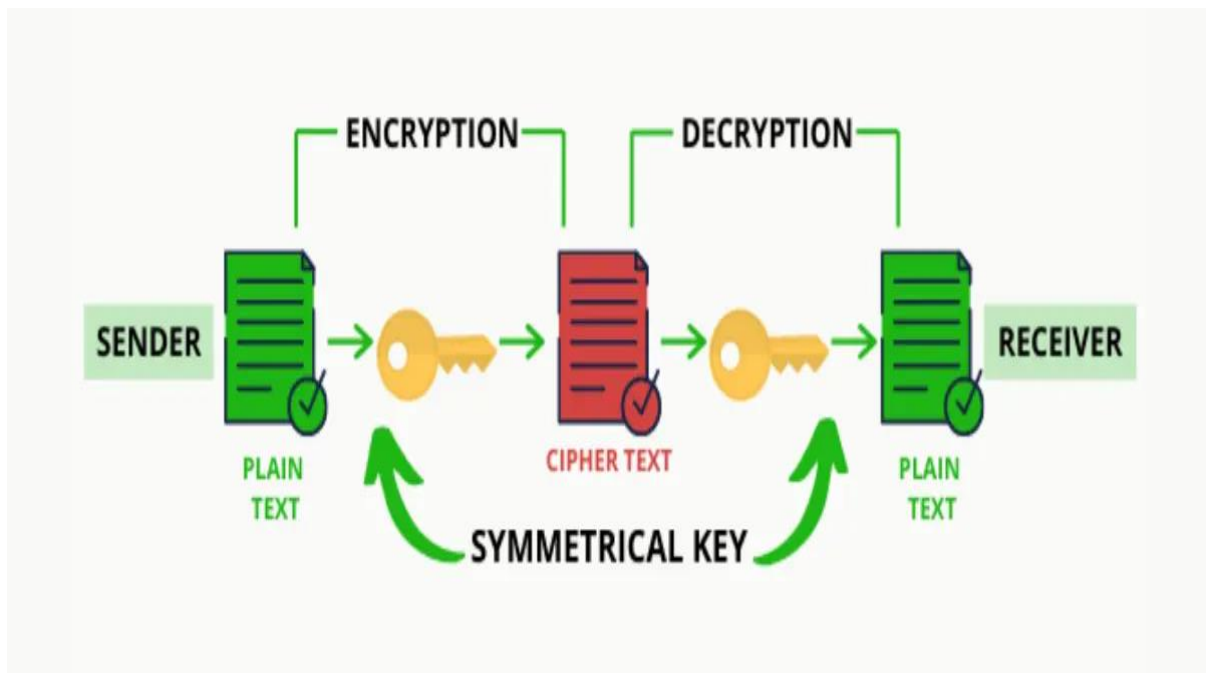
Unit-2 Notes

Cryptocurrency and Bitcoin

Cryptography

[Cryptography](#) is a technique or a set of protocols that secure information from any third party during a process of communication. It is also made up of two Greek terms, Kryptos term meaning “hidden” and Graphein, a term meaning “to write”. Some terminologies related to Cryptography:

- **Encryption:** Conversion of normal text to a random sequence of bits.
- **Key:** Some amount of information is required to get the information of the cryptographic algorithm.
- **Decryption:** The inverse process of encryption, conversion of a Random sequence of bits to plaintext.
- **Cipher:** The mathematical function, i.e. a cryptographic algorithm which is used to convert plaintext to ciphertext(Random sequence of bits).

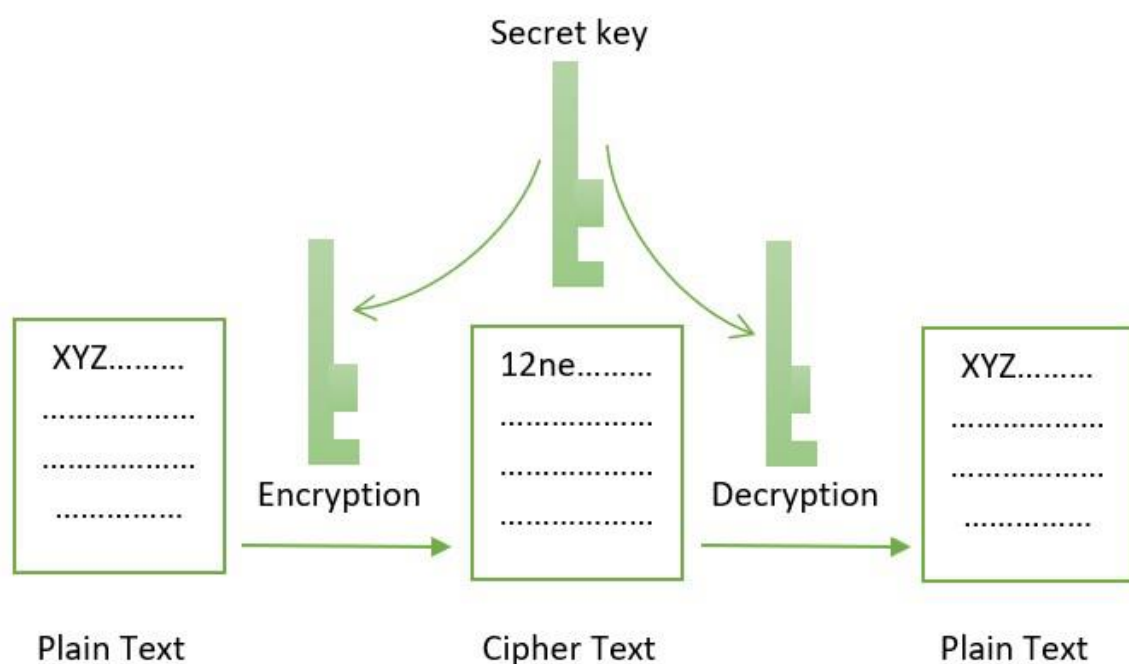


Types of Cryptography in Blockchain

1. **Symmetric-key Encryption**: It focuses on a similar key for encryption as well as decryption. Most importantly, the symmetric key encryption method is also applicable to secure website connections or encryption of data. It is also referred to as secret-key cryptography. The only problem is that the sender and receiver exchange keys in a secure manner. The popular symmetric-key cryptography system is Data Encryption System(DES). The cryptographic algorithm utilizes the key in a cipher to encrypt the data and the data must be accessed. A person entrusted with the secret key can decrypt the data. Examples: AES, DES, etc.

Features:

- It is also known as Secret key cryptography.
- Both parties have the same key to keeping secrets.
- It is suited for bulk encryptions.
- It requires less computational power and faster transfer.



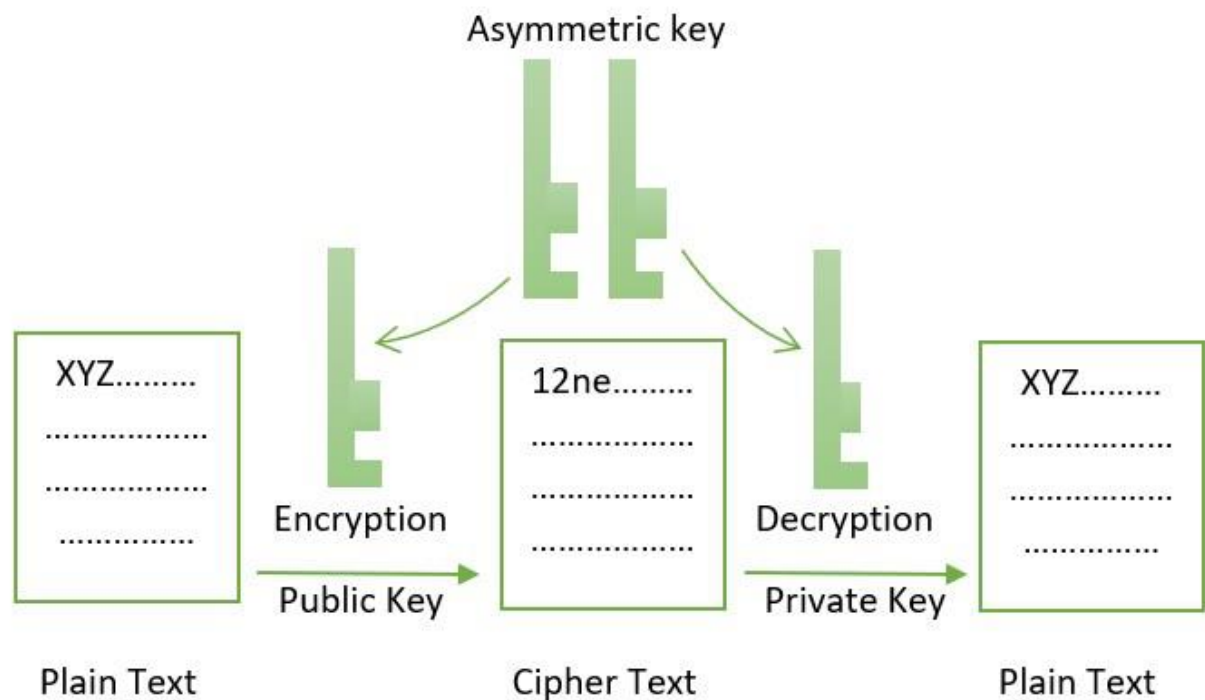
Symmetric Cryptography

2. **Asymmetric-key Encryption**: This cryptographic method uses different keys for the encryption and decryption process. This encryption method uses public and private key methods. This public key method help completely unknown parties to share information between them like email id. private key helps to decrypt the messages and it also helps in the verification of the digital signature. The mathematical relation between the keys is that the private key

cannot be derived from the public key, but the public key can be derived from the private key. **Example:** ECC,DSS etc.

Features:

- It is also known as Public-key cryptography.
- It is often used for sharing secret keys of symmetric cryptography.
- It requires a long processing time for execution.
- Plays a significant role in website server authenticity.



3.Cryptographic Hashing

- The third type of Cryptography is "Hashing" where any plain set of text can be turned into a cryptographic hash (a unique string of data) by using cryptographic algorithms.
- After sending data through a cryptographic hashing, it can't be reversed and hence it holds a distinguished significance other than the above two cryptography techniques.
- The hashing process can also be used to compress the data into a small string of text and hence can help in reducing a large chunk of data.

Benefits of Cryptography in Blockchain

There are a huge number of benefits of cryptography in blockchain some of them are stated below:

- **Encryption:** Cryptography uses asymmetric encryption to ensure that the transaction on their network guards the information and communication against unauthorized revelation and access to information.
- **Immutability:** This feature of cryptography makes it important for blockchain and makes it possible for blocks to get securely linked by other blocks and also to ensure the reliability of data stored in the blockchain, it also ensures that no attacker can derive a valid signature for unposed queries from previous queries and their corresponding signatures.
- **Security:** Cryptography makes the records of transactions easier using encryption of data, and accessing of data using public and private keys. Cryptographic hashing tampering with data is not possible, making blockchain more secure.
- **Scalability:** Cryptography makes the transaction irreversible giving the assurance that all users can rely on the accuracy of the digital ledger. It allows limitless transactions to be recorded securely in the network.
- **Non-repudiation:** The digital signature provides the non-repudiation service to guard against any denial of a message passed by the sender. This benefit can be associated with collision resistance i.e.; since every input value has a unique hash function so there is no clash between the messages that are sent and one message can be easily differentiated from the other.
- **Prevent hackers:** The digital signature prevents hackers from altering the data because if the data changes, the digital signature becomes invalid. With the help of cryptography, it protects the data from hackers and makes cryptography in blockchain unstoppable.

Cryptocurrency – meaning and definition

Cryptocurrency, sometimes called crypto-currency or crypto, is any form of currency that exists digitally or virtually and uses cryptography to secure transactions. Cryptocurrencies don't have a central issuing or regulating authority, instead using a decentralized system to record transactions and issue new units.

What is cryptocurrency?

Cryptocurrency is a digital payment system that doesn't rely on banks to verify transactions. It's a peer-to-peer system that can enable anyone anywhere to send and receive payments. Instead of being physical money carried around and

exchanged in the real world, cryptocurrency payments exist purely as digital entries to an online database describing specific transactions. When you transfer cryptocurrency funds, the transactions are recorded in a public ledger. Cryptocurrency is stored in digital wallets.

Cryptocurrency received its name because it uses encryption to verify transactions. This means advanced coding is involved in storing and transmitting cryptocurrency data between wallets and to public ledgers. The aim of encryption is to provide security and safety.

The first cryptocurrency was Bitcoin, which was founded in 2009 and remains the best known today. Much of the interest in cryptocurrencies is to trade for profit, with speculators at times driving prices skyward.

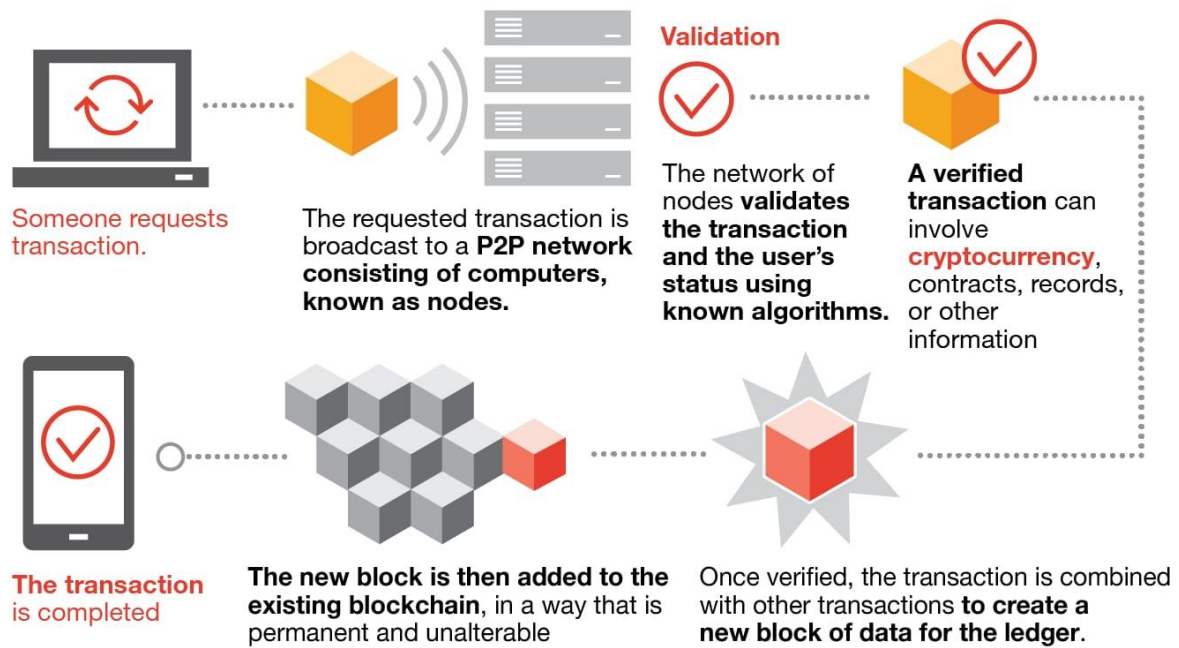
How does cryptocurrency work?

Cryptocurrencies run on a distributed public ledger called blockchain, a record of all transactions updated and held by currency holders.

Units of cryptocurrency are created through a process called mining, which involves using computer power to solve complicated mathematical problems that generate coins. Users can also buy the currencies from brokers, then store and spend those using cryptographic wallets.

If you own cryptocurrency, you don't own anything tangible. What you own is a key that allows you to move a record or a unit of measure from one person to another without a trusted third party.

Although Bitcoin has been around since 2009, cryptocurrencies and applications of blockchain technology are still emerging in financial terms, and more uses are expected in the future. Transactions including bonds, stocks, and other financial assets could eventually be traded using the technology.



Features of Cryptocurrency

There are various features of Cryptocurrency. These features made cryptocurrency different from our regular fiat currency, added more security to the money and is hence increasing the popularity of the cryptocurrency. Here are some features of cryptocurrency.

- **Decentralization:** Cryptocurrency is not regulated or controlled by any authorities or any bank, it shares networks and completely works based on Blockchain Technology. It is instead collectively retained control by all users. It is immutable meaning all its transactions are not reversible.
- **Accountable and Transparent:** Transactions done at block level are highly transparent and could be seen by everyone. It helps in preventing any frauds. Any transaction once done cannot be traced back or it is not reversible. Hence, at most care need to be taken while doing any transaction
- **Digitization:** Cryptocurrency exists only in digital form. No physical notes or coins exist, hence they could be easily transferred via digital platforms. It makes the user more secured as they could not be theft by anyone. The user also need not maintain physical space in order to store currency.
- **Global Transactions:** Cryptocurrencies can be transferred digitally to anyone and anywhere without any border restrictions. Since, it is digitized, transactions could be done at a single click at the comfort of the user. Most of the countries in the world use these cryptocurrency transactions.
- **Volatile Price:** Cryptocurrencies are very high price volatile within short period of time, at the same time they are considered risky. It is mostly

believed that it brings high return on investment within very short period of time. Hence, many would invest in it.

- **Security:** Cryptocurrencies follow cryptographic algorithms, which make them more secure and makes the hackers more difficult in manipulating it . That is why many users show interest in investing in cryptocurrency.

Types of Cryptocurrency

1. Bitcoin (BTC)

Bitcoin was the world's first cryptocurrency, with its origins dating back to a white paper published in 2008, and remains the best-known type of crypto. It functions on its own [blockchain](#), with transactions verified (and new Bitcoins created, up to a fixed cap) by an army of decentralized miners.

2.Ethereum

Developed in 2015, Ethereum is a blockchain platform with its own cryptocurrency, called Ether (ETH) or Ethereum. It is the most popular cryptocurrency after Bitcoin.

3. Tether (USDT)

Tether is a type of *stablecoin*, designed to have a less-volatile price by being linked to an external asset. In this case, each coin is backed by an equivalent number of US dollars, which keeps it from experiencing the same kind of pricing volatility that other cryptocurrencies face.

4. Binance Coin (BNB)

Binance Coin is native to Binance, the world's largest cryptocurrency exchange as of 2021. Transaction fees for this exchange are reduced for users who choose to pay in BNB.

5.USD Coin (USDC)

Much like Tether, USD Coin is a stablecoin connected to the US dollar that cannot be mined. However, unlike Tether, USD Coin has more transparent funding and better auditing processes.

XRP

XRP is the [native token](#) for the XRP Ledger, created as a payment system by Ripple in 2012. The XRP Ledger uses a consensus mechanism called the XRP

Ledger Consensus Protocol, which doesn't use proof-of-work or proof-of-stake for consensus and validation. Instead, client applications sign and send transactions to the ledger servers. The servers then compare the transactions and conclude whether they are candidates for entry into the ledger.

Cryptocurrency Use Cases

1. Decentralised digital currency

The first and foremost use case of cryptocurrencies is that it has provided the world with a decentralised digital currency. It means that this currency has no government or other third-party interference, and banks or other financial organisations cannot exercise any control over this digital coin.

2. Crypto banking

Banking barons like Barclays, JP Morgan, Goldman Sachs, and others have tailored their banking services to manage cryptocurrencies. This has led to the emergence of a system called crypto banking. These banks have begun offering crypto interest accounts and savings accounts.

Traditional banks hold cash, stocks and other financial assets for their users. Similarly, crypto banks hold their clients' crypto assets and traditional fiat currencies. Do note that India has to do a lot of catching up in this regard.

3. Crypto staking

Crypto staking is one of the best ways to earn passive income from crypto assets. Proof-of-Stake (PoS) blockchains allow you to stake your cryptocurrency and participate in its consensus mechanism. In exchange, you can validate a block on that chain and earn rewards.

The cryptocurrencies you can stake are:

Ethereum (ETH)

Solana (SOL)

Cardano (ADA)

Avalanche (AVAX)

Luna (LUNA)

Polkadot (DOT)

You can safely rely on PoS blockchains for investing in your crypto. As the network forces its participants to lock away a certain portion of its local tokens, it safeguards against any malicious activity on that blockchain.

4. Asset tokenisation

Cryptocurrencies allow you to tokenise physical assets and link them to digital tokens. Thus, you can tokenise commodities, real estate, art, stocks, copyrights, etc. This process of asset digitisation improves the market liquidity of various real-world assets. Asset tokenisation would allow increased participation of investors who would have otherwise been unable to invest due to inadequate cash.

Furthermore, by dividing the ownership of items into several pieces, investors can gain a proportion of the value of assets they hold. Asset tokenisation would be a popular use case in our country, especially in Tier-2 and Tier-3 cities.

5. Online payments

Cryptocurrencies have revolutionised international payments through their high transaction speeds, decentralised nature and low transaction costs. Your crypto payments are anonymous, and no one can track you in real life based on your online activities.

Moreover, you can also use your crypto assets at merchants that accept crypto as a form of payment. Leading enterprises like Mastercard, Pepsi, Starbucks, Tesla, Visa, McDonald's, etc., accept crypto from their customers as a valid payment method.

6. On-chain governance

Cryptocurrencies offer a more refined way to implement new policies on a network. Blockchains have DAOs (Decentralised Autonomous Organisations), allowing you to stake crypto coins in exchange for voting rights.

Blockchain developers call it Governance Token. On a blockchain, the community can distribute governing power among its stakeholder via these tokens. This is called On-Chain Governance. By owning governance tokens, you get the authority to alter a network's protocol.

7. Crypto gaming

The crypto gaming industry has seen a boom in recent years. Gamers worldwide are earning in-game items in the form of NFTs and trading and selling them based on their demand.

These P2E games can be an excellent way to utilise your crypto assets and gain endless hours of fun and NFTs in the form of rewards. Furthermore, you can earn crypto as a reward by playing these games.

There are a lot of gamers in the Philippines who earn a daily wage from playing Axie Infinity, a popular NFT game.

8. Smart contracts

Smart contracts are lines of code that execute a transaction if a given set of terms and conditions are met among the concerned parties. You can automate various actions and functionalities using smart contracts.

The best part about them is that there are no interruptions in the execution of a process when the underlying terms are met. Furthermore, no third parties will have the authority to change or modify the decision or agreement. As smart contracts are blockchain-based, all the data it generates is immutable, and no entity can update or alter it in any way.

Smart contracts have an increasing number of use cases in the fields of Governance, Finance, Health Care, Insurance etc.

9. Store of value

You may have heard crypto investors call Bitcoin “digital gold”. This is because cryptocurrencies can store and transfer value over time. They work on a supply mechanism that restricts inflation. As the value of crypto increases with a rise in demand, it can serve as an excellent investment opportunity over time.

Some experts even predict that Bitcoin will take the market share away from gold in the coming time.

10. Borrowing and lending

You can use your crypto coins to lend or borrow funds via Decentralised Finance (DeFi) applications. As crypto has no borders, you can borrow or lend capital from any part of the world using your crypto assets. These DeFi apps also reduce the time and the huge amount of paperwork you need to complete while getting a loan from a traditional bank.

11. NFTs

NFTs, or non-fungible tokens, are one of the best investments you can make with cryptocurrencies. They are unique cryptographic tokens that exist on a blockchain that individuals cannot replicate. These digital tokens can represent a variety of items. They are:

Artwork

Real estate

Digital artwork

Sports cards

Rare items and much more

How to buy cryptocurrency

You may be wondering how to buy cryptocurrency safely. There are typically three steps involved. These are:

Step 1: Choosing a platform

The first step is deciding which platform to use. Generally, you can choose between a traditional broker or dedicated cryptocurrency exchange:

- **Traditional brokers.** These are online brokers who offer ways to buy and sell cryptocurrency, as well as other financial assets like stocks, bonds, and ETFs. These platforms tend to offer lower trading costs but fewer crypto features.
- **Cryptocurrency exchanges.** There are many cryptocurrency exchanges to choose from, each offering different cryptocurrencies, wallet storage, interest-bearing account options, and more. Many exchanges charge asset-based fees.

When comparing different platforms, consider which cryptocurrencies are on offer, what fees they charge, their security features, storage and withdrawal options, and any educational resources.

Step 2: Funding your account

Once you have chosen your platform, the next step is to fund your account so you can begin trading. Most crypto exchanges allow users to purchase crypto using fiat (i.e., government-issued) currencies such as the US Dollar, the British Pound, or the Euro using their debit or credit cards – although this varies by platform.

Crypto purchases with credit cards are considered risky, and some exchanges don't support them. Some credit card companies don't allow crypto transactions either. This is because cryptocurrencies are highly volatile, and it is not advisable to risk going into debt — or potentially paying high credit card transaction fees — for certain assets.

Some platforms will also accept ACH transfers and wire transfers. The accepted payment methods and time taken for deposits or withdrawals differ per platform. Equally, the time taken for deposits to clear varies by payment method.

An important factor to consider is fees. These include potential deposit and withdrawal transaction fees plus trading fees. Fees will vary by payment method and platform, which is something to research at the outset.

Step 3: Placing an order

You can place an order via your broker's or exchange's web or mobile platform. If you are planning to buy cryptocurrencies, you can do so by selecting "buy," choosing the order type, entering the amount of cryptocurrencies you want to purchase, and confirming the order. The same process applies to "sell" orders.

How to store cryptocurrency

Once you have purchased cryptocurrency, you need to store it safely to protect it from hacks or theft. Usually, cryptocurrency is stored in crypto wallets, which are physical devices or online software used to store the private keys to your cryptocurrencies securely. Some exchanges provide wallet services, making it easy for you to store directly through the platform. However, not all exchanges or brokers automatically provide wallet services for you.

There are different wallet providers to choose from. The terms “hot wallet” and “cold wallet” are used:

- **Hot wallet storage:** "hot wallets" refer to crypto storage that uses online software to protect the private keys to your assets.
- **Cold wallet storage:** Unlike hot wallets, cold wallets (also known as hardware wallets) rely on offline electronic devices to securely store your private keys.

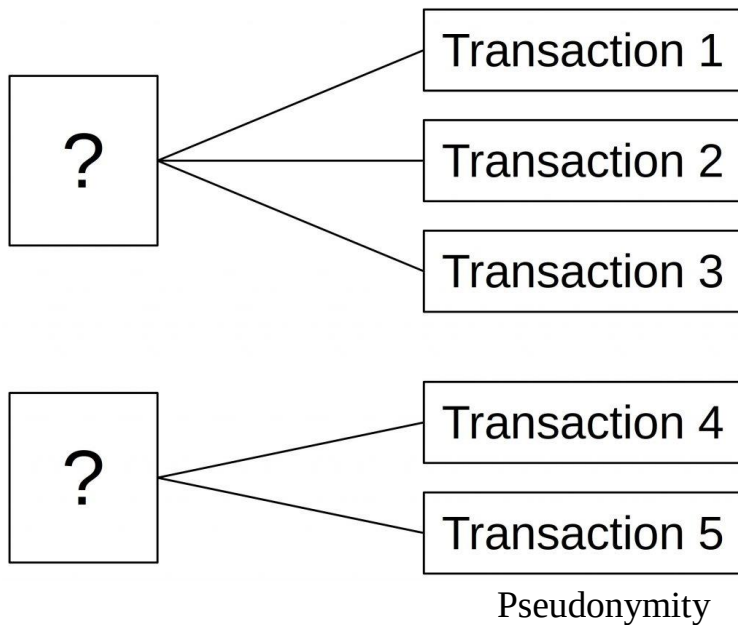
Typically, cold wallets tend to charge fees, while hot wallets don't.

Anonymity and Pseudonymity in cryptocurrencies

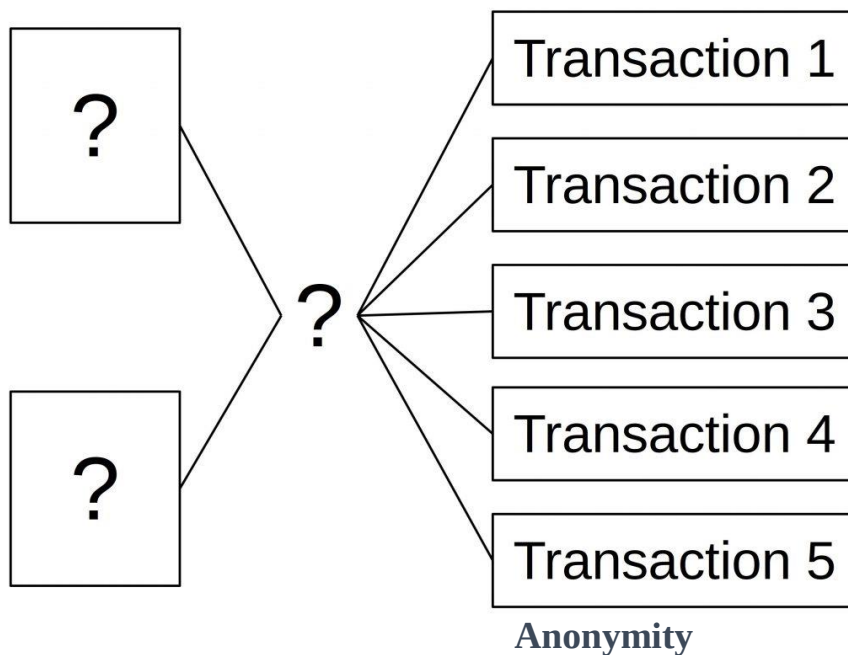
Anonymity means that the identity of a person is unknown and its actions are not trackable.

Pseudonymity, however, is a much weaker form. It means that a person's identity is unknown, but it is possible to assign actions to the same person.

The first figure shows a situation where the person's identity is unknown. But all the actions the person made can be linked to this person.



The second figure shows a situation where a person's identity is unknown and it is impossible to link any transaction to a specific person.



Blockchains like Bitcoin or Ethereum are pseudonymous instead of anonymous. The address is a pseudonym for the person holding the corresponding private key. Every action done with this address can be traced back to this person, although its identity might be unknown. And since all transactions are stored in the blockchain, it is possible for everybody to trace them back, even though this transaction lies long in the past.

There are so-called privacy coins like Monero or ZCash that allow for real anonymity through cryptography. Here, transactions cannot be linked to individual addresses.

Digital Signatures

- A digital signature is a mathematical scheme that is used to verify the integrity and authenticity of digital messages and documents.
- It may be considered as a digital version of the handwritten signature or stamped seal. The digital signatures use asymmetric cryptography i.e. also known as public key cryptography.
- Digital signatures use asymmetric key cryptography. [Asymmetric key cryptography](#) also known as public key cryptography uses public and private keys to encrypt and decrypt data.
- The public key can be shared with anyone.
- The private key is the secret key that is kept a secret.
- It can be summarized as a digital signature a code that is attached to the message sent on the network. This code acts as proof that the message hasn't been tampered with along its way from sender to receiver.

Why are Digital Signatures Important?

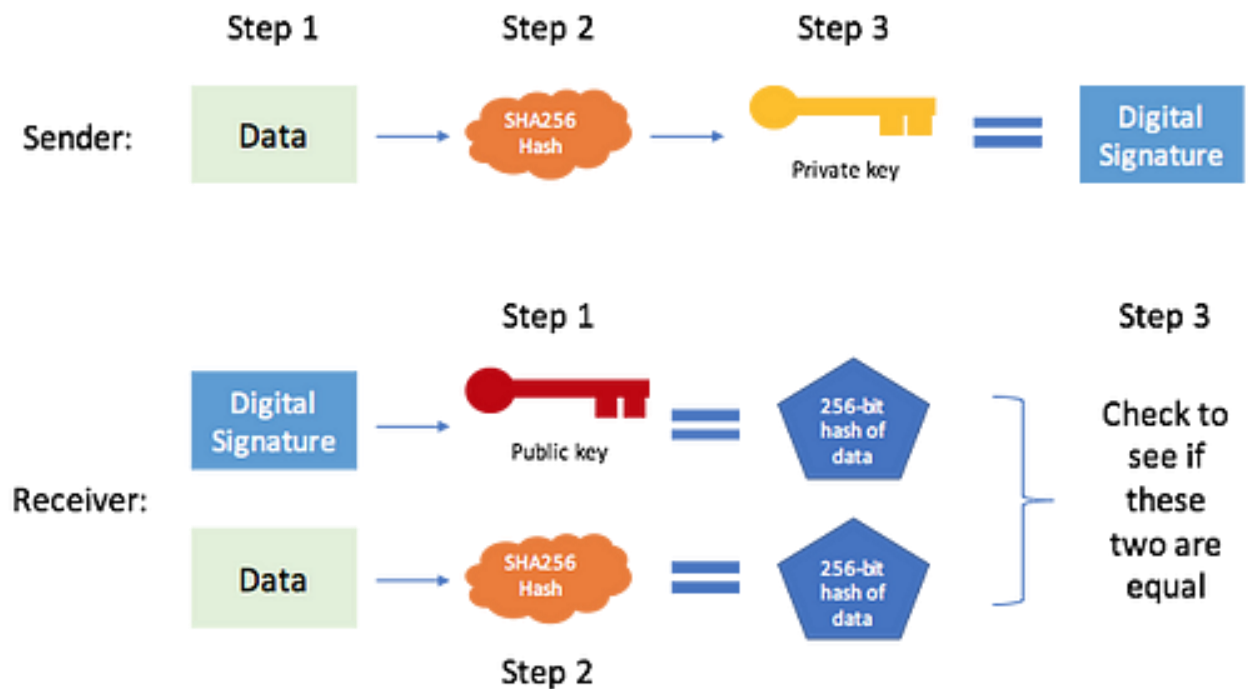
- Digital signatures are important to achieve three results: Data integrity, authenticity, and non-repudiation
- **Data Integrity:** It is preserved by using the hash function in signing and verifying algorithms. Any change in the message will produce a completely different signature.
- **Authenticity:** The message is verified using the public key of the sender. When Alice sends a message to Bob.
- **Message Nonrepudiation:** Once the signature is generated, Alice cannot deny having signed it in the future, unless Alice's private key is compromised.

Let's understand the above concept using an example:

- Alice decides to send a message to Bob.
- Alice creates the hash value of the document.
- Alice uses her private key to encrypt the hash value.
- Alice sends the document along with the encrypted hash value to Bob.
- When Bob receives the message, he will use Alice's public key to decrypt the received hash value.
- Bob will also generate the hash value of the message received.

- Bob will match the two hash values and if the values match then Bob will be sure that the message has not been tampered with. If the values don't match then it is confirmed that the message has been tampered with.

Bitcoin uses a process called Digital Signatures to validate authenticity of a transaction and the sender of that transaction. You use the public/private key pairs to digitally sign data in order to prove that the data has not been tampered with and validate that the data must have been sent by the person who owns the private key to the given public key.



Step 1: The sender has some data that he/she would like to send to the receiver. The receiver wants to ensure that the data he/she receives from the sender has not been tampered and must have come from the sender and nobody else.

Step 2: The sender collects the data to be sent. Once finalized, the sender uses the SHA256 hash function to hash the data into a 256-bit number.

Step 3: The sender then signs the 256-bit number with his/her private key encrypting the 256-bit number into what's called a Digital Signature. Now the sender sends the data, the Digital Signature, and his/her public key to the receiver (remember, you can't use the public key to figure out the corresponding private key so it is ok to share).

The receiver must authenticate that the data sent was not altered and must have been sent by the person who has the corresponding private key to the shared public key.

Step 1: The receiver takes the Digital Signature and decrypts it with the received public key to get a 256-bit number. Applying the public key to the Digital Signature 'reverses' the sender's step 3 above.

Step 2: Then, the receiver takes the received data and applies the SHA256 hash to it to get a 256-bit number. This is the same thing as the sender's step 1 and 2.

Step 3: The receiver then checks to make sure the two 256-bit numbers are equal. If false, then somebody has tampered with the data or provided a public key that does not correspond to the private key of the sender. If true, the receiver knows the data is good to go.

Why are Digital Signatures Important?

Digital signatures are important to achieve three results: Data integrity, authenticity, and non-repudiation.

1. **Data Integrity:** It is preserved by using the hash function in signing and verifying algorithms. Any change in the message will produce a completely different signature. This way Bob can verify that the message sent by Alice was not modified along its way.
2. **Authenticity:** The message is verified using the public key of the sender. When Alice sends a message to Bob. Bob uses the public key of Alice for verification and Alice's public key cannot create the same signature as Kev's private key.
3. **Message Nonrepudiation:** Once the signature is generated, Alice cannot deny having signed it in the future, unless Alice's private key is compromised.
- 4.

What is Bitcoin?

Satoshi Nakamoto introduced the bitcoin in the year 2008. Bitcoin is a cryptocurrency(virtual currency), or a **digital currency** that uses rules of cryptography for regulation and generation of units of currency. A Bitcoin fell under the scope of [cryptocurrency](#) and became the first and most valuable among them. It is commonly called **decentralized digital currency**.

A bitcoin is a type of digital assets which can be bought, sold, and transfer between the two parties securely over the internet. Bitcoin can be used to store values much like fine gold, silver, and some other type of investments. We can also use bitcoin to buy products and services as well as make payments and exchange values electronically.

A bitcoin is different from other traditional currencies such as **Dollar**, **Pound**, and **Euro**, which can also be used to buy things and exchange values electronically. There are no physical coins for bitcoins or paper bills. When you send bitcoin to someone or used bitcoin to buy anything, you don't need to use a bank, a credit card, or any other third-party. Instead, you can simply send bitcoin directly to another party over the internet with securely and almost instantly.

Basic Components of Bitcoin

There are four basic components of bitcoin

1. **Software:** Bitcoin is, at its heart, a piece of software that defines what a bitcoin is and how it is transmitted. Checking of validity or who is allowed or not allowed to be within bitcoin etc. Type of the regulations of a legitimate bitcoin is established via it.

2. **Cryptography:** Cryptography and bitcoin as a cryptocurrency are at the heart of the software. Bitcoin regulates both the transfer of bitcoin between parties and the production of new bitcoin units using encryption.
3. **3. Hardware:** Cryptography demands a lot of hardware to run and solve. This gear has been created specifically for mining, i.e. detecting Nonce to validate blocks and hashes to accomplish a simple activity on the bitcoin blockchain, a lot of CPU power is required.
4. **4. Miners(Gaming Theory):** Game theory studies rational decision-making behavior in humans. Game theory allows interactions between two or more players in a system where the participant's outcome is based on the actions of the others.

Advantages of Bitcoin

1.Ease of transactions

- The original Bitcoin whitepaper defines it as a digital peer-to-peer currency which makes instantaneous transactions. Unlike payment networks like PayPal and Visa, Bitcoins incur very low transaction surcharges

2. Anonymity and decentralisation

- Alphanumeric cloaks hide Bitcoin users' identities and prevent illegitimate access. Although transactions are visible through connecting data points, Bitcoins enable a pseudonymous account that can safeguard user information.

3. Value appreciation

- Owing to the limited supply of Bitcoins since its beginning and its increasing usage, Bitcoins have appreciated. Unlike fiat money, the value of Bitcoins fluctuates with every transaction

4.Security and free from market forces

- Unlike fiat money transactions prone to cyber-attacks and fraudulent activities, Bitcoins are encrypted and immune to seizure. Every Bitcoin transaction is visible on an openly distributed ledger, making unauthorised changes difficult.

5.Tax-free and zero transaction costs

- Most countries do not levy taxes on Bitcoin returns. Since third-party applications cannot intercept such transactions, it isn't easy to implement a stable taxation policy

Disadvantages of Bitcoins

1. Volatility

- The volatile nature of cryptocurrencies depends on factors like limited supply, increasing market demand, investor sentiment, etc. The limited supply and growing demand make its value very susceptible to fluctuations

2. Absence of regulations

- Although potential investors consider the absence of government regulations a determining factor, its decentralisation makes it devoid of legal protection.

3. Irreversibility and limited usage

- The irreversibility of Bitcoins adds to their unregulated and anonymous nature. Any accidental payment cannot be traced and therefore is risky.

4. Uncertain future

- While several nations like El Salvador have accepted Bitcoin as a regular payment mode, many countries have barred its usage. The economic shock posed by the Covid-19 pandemic has forced countries like Qatar, China, Turkey, North Macedonia, Egypt, Iraq, and Bangladesh to get into Bitcoins.

5. Technical flaws and a deflationary effect

- Since Bitcoins is a relatively newer concept, the blockchain network has innumerable flaws and loopholes

How Bitcoin Works?

When you send an email to another person, you just type an email address and can communicate directly to that person. It is the same thing when you send an instant message. This type of communication between two parties is commonly known as Peer-to-Peer communication.

Whenever you want to transfer money to someone over the internet, you need to use a service of third-party such as banks, a credit card, a PayPal, or some other type of money transfer services. The reason for using third-party is to ensure that you are transferring that money. In other words, you need to be able to verify that both parties have done what they need to do in real exchange.

For example, Suppose you click on a photo that you want to send it to another person, so you can simply attach that photo to an email, type the receiver email address and send it. The other person will receive the photo, and you think it would end, but it is not. Now, we have two copies of photo, one is a simple email, and another is an original file which is still on my computer. Here, we send the copy of the file of the photo, not the original file. This issue is commonly known as the double-spend problem.



The double-spend problem provides a challenge to determine whether a transaction is real or not. How you can send a bitcoin to someone over the internet without needing a bank or some other institution to certify the transfer took place. The answer arises in a global network of thousands of computers called a Bitcoin Network and a special type of decentralized ledger technology called **blockchain**.

In Bitcoin, all the information related to the transaction is captured securely by using maths, protected cryptographically, and the data is stored and verified across the entire network of computers. In other words, instead of having a centralized database of the third-party such as banks to certify the transaction took place. Bitcoin uses [blockchain](#) technology across a decentralized network of computers to securely verify, confirm and record each transaction. Since data is stored in a decentralized manner across a wide network, there is no single point of failure. This makes blockchain more secure and less prone to fraud, tampering or general system failure than keeping them in a single centralized location.

Digital Keys and Addresses

Digital keys play a central role in the bitcoin network because they act as proof of ownership for the digital currency, and are necessary when signing digital transactions. Public-key cryptography is the science behind the keys, the encryption process basically uses two mathematically related, but not identical, keys – a public key and a private key, and each key performs a specific function.

What's important to understand is that these digital keys are created and managed by digital wallets. Each wallet runs a digital (public) address, similar to the beneficiary name on a traditional paper cheque. It consists of a public key (similar to a credit card number) and a private key (similar to a credit card PIN number). In the conventional, non-digital world, the safety of your underlying funds is not compromised by having the beneficiary name printed on a cheque or entering full credit card numbers when making an online purchase. Similarly with public key cryptography, the public key and digital address are shared and exposed. However the private key, is just like your PIN on a bank card, so it should never be shared. Private keys should be kept as secret and safe as PIN numbers because they authorise the spending of your money.

Private Key

A private key is a randomly generated 256-bit number. Each bit (short for **binary digit**) is represented by one of two binary values, a 0 or a 1.



The following is an example of a random private key as a series of 256 bits:

```
1110111100100011010110101010110011111001000011011001111101001010
1010110111011000110010010010111001001011001001010110001011100001
1101100111101011100101111111000011011111100110111010001110110101
0000100000100101100001110011100111001011000000010011110110110010
```

To wrap your head around the enormous amount of possible 256-bit combinations, just think of it as approximately 10^{77} . Now compare that to the number of the stars in the universe, which is around 10^{24} . Pretty huge, right?

In the world of mathematics and computer science, we put 1s and 0s into groups of 4 bits, known as half bytes. Thus, if we take the above example of a private key and format it in groups of half bytes, it would look as follows:

```
1110 1111 0010 0011 0101 1010 1010 1100 1111 1001 0000 1101 1001 1111
0100 1010 1010 1101 1101 1000 1100 1001 0010 1110 0100 1011 0010 0101
0110 0010 1110 0001 1101 1001 1110 1011 1001 0111 1111 0000 1101 1111
1001 1011 1010 0011 1011 0101 0000 1000 0010 0101 1000 0111 0011 1001
1100 1011 0000 0001 0011 1101 1011 0010.
```

As there are a total of 16 possible 4-bit combinations, each half byte is made to correspond to a decimal number between 0 and 15, as per the table below:



Decimal Number	Half Byte			
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
10	1	0	1	0
11	1	0	1	1
12	1	1	0	0
13	1	1	0	1
14	1	1	1	0
15	1	1	1	1

So, using the above table, we can re-write our private key example by mapping each half byte to the corresponding decimal number. The result looks like so:

14 15 2 3 5 10 10 15 12 15 9 0 13 9 15 4 10 10 13 13 8 12 9 2 14 4 11 2 5 6 2
 14 1 13 9 14 11 9 7 15 0 13 15 9 11 10 3 11 5 0 8 2 5 8 7 3 12 11 0 1 3 13 11 2

Next to the binary system that has 2 digits (0 or 1), and the decimal system that consists of 10 digits from 0 to 9, there is also the hexadecimal system which consists of 16 alphanumeric characters, ordered from 0 to F. These are 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, and F.

Decimal Number	Half Byte				Hexadecimal
0	0	0	0	0	0
1	0	0	0	1	1
2	0	0	1	0	2
3	0	0	1	1	3
4	0	1	0	0	4
5	0	1	0	1	5
6	0	1	1	0	6
7	0	1	1	1	7
8	1	0	0	0	8
9	1	0	0	1	9
10	1	0	1	0	A
11	1	0	1	1	B
12	1	1	0	0	C
13	1	1	0	1	D
14	1	1	1	0	E
15	1	1	1	1	F

Keeping all three systems in mind, our original 256-bit private key converts to a hexadecimal number as follows:

ef235aacf90d9f4aadd8c92e4b2562e1d9eb97f0df9ba3b508258739cb013db2

Private keys appear as 51 alphanumeric characters when the so-called Base58 encoding system is used. Base58 is a binary-to-text encoding structure which includes both upper and lower case characters of the English alphabet, as well as numbers 1 to 9. In order to avoid any confusion and mistakes with similar looking characters in typing private keys, the number zero (0), the lower and upper case letter o, capital i (I) and lower case L (l), have all been omitted.

Private keys encoded with the Base58 format always start with 5H, 5J or 5K. For example,

5HpHagT65TZzG1H3CSu63k8Dbpv8s5ip4nEB3kEsreAvUcVfH34

Public Keys

Public keys are created from private keys using a mathematical concept known as “elliptic curve multiplication”, which can generate a unique public key from a private key but more importantly, not the other way around. In this way, a private key is used to digitally sign (or authenticate) a transaction to spend a number of bitcoins. By publicising the signature and the public key, anyone can

infer that the transaction is valid without needing to actually see the private key. In other words, they can see that:

- The transaction was indeed sent by the private key holder, i.e. the legitimate owner of the bitcoins, and
- At the time of the transaction, the owner owned the amount of bitcoins that appears on the transaction

Public keys can be either 520 bits (65 bytes) long, or 264 bits (33 bytes) long if the key is compressed.

Bitcoin Address

To get to the bitcoin address, the public key is fed into a cryptographic algorithm known as SHA (Secure Hash Algorithm), designed by the United States National Security Agency (NSA). The SHA-256 algorithm takes an input string of any size and converts it to exactly 256 bits. After this, the resulting 256-bit long string is fed again to the SHA algorithm, and more specifically to RIPEMD-160, another family of cryptographic algorithms.

The end result is a 40 characters long Bitcoin Address:



Bitcoin Transaction

- Bitcoin transaction means sending bitcoin from one person to the other in the secured blockchain network.
- These are messages that are digitally signed using cryptography and are verified by the miners that are present in the blockchain network.
- The miner is the person who solves mathematical puzzles(also called proof of work) to validate the transaction.
- Anyone with mining hardware and high processing power can take part in this. Numerous miners take part simultaneously to solve the complex mathematical puzzle, the one who solves it first, wins 12.5 bitcoin as a reward.
- Miner verifies the transactions(after solving the puzzle) and then adds the block to the blockchain when confirmed
- The transaction input is the bitcoin address from which the money was sent, and the transaction output is the bitcoin address to which the money was sent.
- Generally, a bitcoin transaction takes 10 to 20 minutes to confirm any transactions. if network congestion takes place, then time might take even 60 minutes.

Example: Suppose Hari wants to send 5 BTC(bitcoins) to Seniorita.

- Input- It consists of the bitcoin address from which hari got his bitcoins previously.
- Amount- This deals the specific amount of money that hari wishes to send to seniorita, in this case, its 5BTC.
- Output- This is the bitcoin address of seniorita which hari needs to know in order to transfer the bitcoins to her wallet.
- Bitcoin address of seniorita is also known as public key.

What is Mining

Bitcoin mining is a computation-intensive process that uses complicated computer code to generate a secure cryptographic system. The bitcoin miner is the person who solves mathematical puzzles(also called proof of work) to validate the transaction. Anyone with mining hardware and computing power can take part in this. Numerous miners take part simultaneously to solve the complex mathematical puzzle, the one who solves it first, wins 6.25 bitcoin as a part of the reward. Miner verifies the transactions(after solving the puzzle) and then adds the block to the blockchain when confirmed. The blockchain contains the history of every transaction that has taken place in the blockchain network.

Once the minor adds the block to the blockchain, bitcoins are then transferred which were associated with the transaction.

For the miners to earn rewards from verifying the bitcoin Transactions, two things must be ensured:

1. The miners must verify the one-megabyte size of the transaction.
2. For the addition of a new block of transaction in the blockchain, miners must have the ability to solve complex computational maths problems called proof for work by finding a 64-bit hexadecimal hash value.

Why Do Bitcoin Needs To Be Mined?

Bitcoin is a digital currency where there are chances of copying, counterfeiting, or double-spending the same coin more than once. Mining solves these problems by making the above illicit activities extremely expensive and resource-intensive. Thus, it can be concluded that it is more beneficial and cost-effective to join the network as a miner than to try to undermine it.

Why Does Bitcoin Needs Miners?

Bitcoin miners are very essential for the smooth functioning of the bitcoin network for the following reasons:

- Miners' job is just like auditors i.e. to verify the legitimacy of the bitcoin transactions.
- Miners help to prevent the [double-spending problem](#).
- Miners are minting the currency. In the absence of miners, Bitcoin as the network would still exist and be usable but there would be no additional bitcoin.

Why Mine Bitcoins?

There are several pros of mining a bitcoin:

- Mining bitcoin helps support the Bitcoin ecosystem.
- Bitcoin mining helps miners to earn rewards in form of bitcoins.
- It is the only way to release new cryptocurrencies into circulation.
- It is used to check counterfeiting and double spending.

How Does Bitcoin Mining Work?

The nodes of the blockchain network are based on the concept that no one in the network can be trusted. Proof of work is accepted by nodes to validate any transaction. Proof of work involves doing hefty calculations to find a 32-bit hash value called nonce to solve the mathematical puzzle. The miners create new blocks by abiding by the fact that the transaction volume must be less than

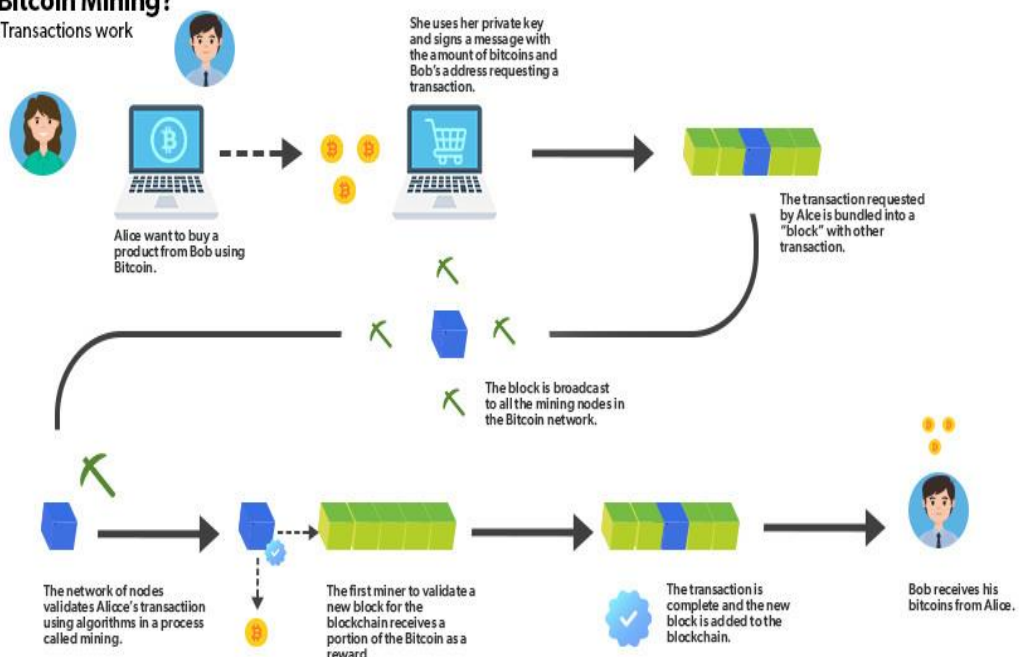
21 million. 21 million is the total number of bitcoins that can be generated. The verified transaction gets a unique identification code and is linked with the previous verified transaction.

Let's understand this with the help of an example-

- Suppose Alice wants to transfer 10 BTC to Bob.
- Now the transaction data of A is shared with the miners from the memory pool. A memory pool is a place where an unconfirmed or unverified transaction waits for its confirmation.
- Miners start competing with themselves to solve the mathematical riddle in order to validate and verify the transaction using proof of work.
- The miner who solves the problem first shares his result with other nodes(miners).
- Once maximum nodes agree with the solution, the transaction block is verified and is then added to the blockchain.
- At the same time, the miner who solved the puzzle gets a reward of 6.25 bitcoins.
- Now, after the addition of the transaction block, the 10 BTC associated with the transaction data is transferred to Bob from Alice.

What is Bitcoin Mining?

How Bitcoin Transactions work



What is Double Spending in Blockchain?

Blockchain is a list of blocks. Each block comprises some information associated with some hash. Blockchain is used nowadays widely for transactions. It is an immutable, distributed, and decentralized ledger. The working of Blockchain is as follows. Suppose a user wants to make a transaction. A block is created and sent to other users. Users validate the block and the transaction gets executed. The block is added and the users get incentives.

Understanding Double Spending

Although Blockchain is secured, still it has some loopholes. Hackers or malicious users take advantage of these loopholes to perform their activities.

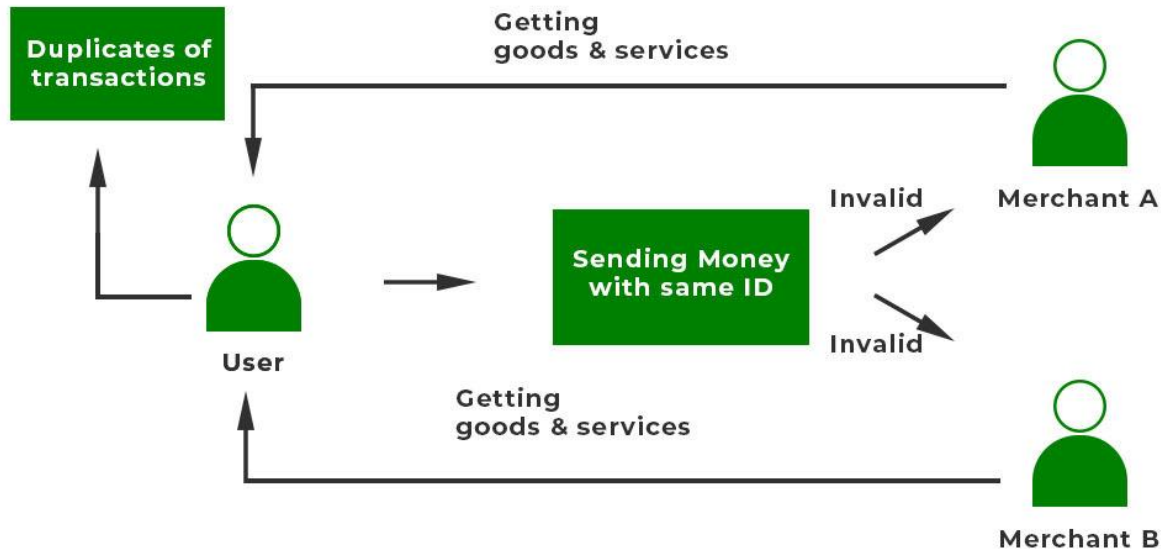
- [Double spending](#) means the expenditure of the same digital currency twice or more to avail the multiple services. It is a technical flaw that allows users to duplicate money.
- Since digital currencies are nothing but files, a malicious user can create multiple copies of the same currency file and can use it in multiple places.
- This issue can also occur if there is an alteration in the network or copies of the currency are only used and not the original one.
- There are also double spends that allow hackers to reverse transactions so that transaction happens two times.
- By doing this, the user loses money two times one for the fake block created by the hacker and for the original block as well.
- The hacker gets incentives as well for the fake blocks that have been mined and confirmed.

How Does Double Spending Happen?

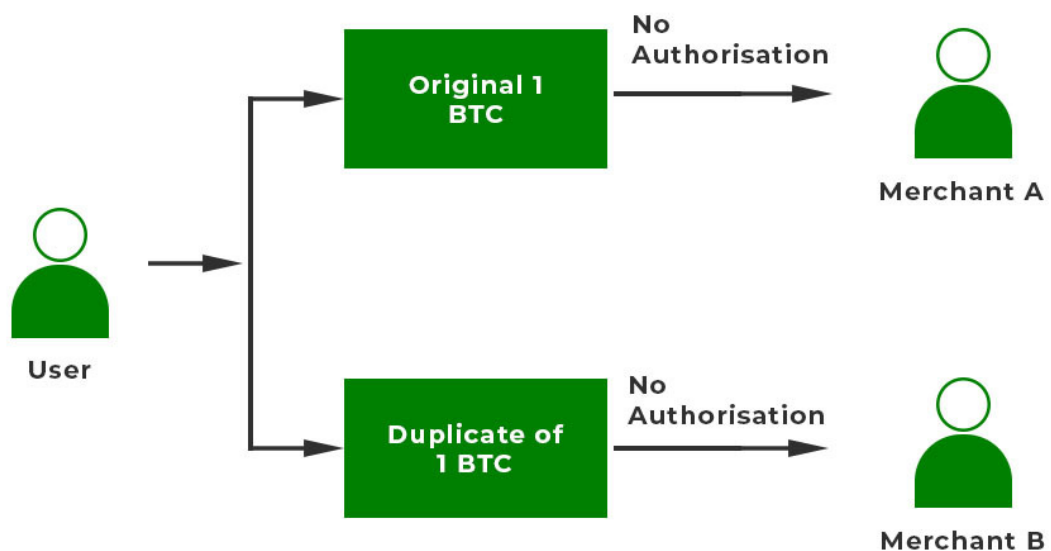
Double spending can never arise physically. It can happen in online transactions. This mostly occurs when there is no authority to verify the transaction. It can also happen if the user's wallet is not secured. Suppose a user wants to avail of services from Merchant 'A' and Merchant 'B'.

- The user first made a digital transaction with Merchant 'A'.
- The copy of the cryptocurrency is stored on the user's computer.
- So the user uses the same cryptocurrency to pay Merchant 'B'.
- Now both the merchants have the illusion that the money has been credited since the transactions were not confirmed by the miners.

This is the case of double spending.



Example: Suppose a user has 1 BTC. He/She wants to avail of services from merchant A and merchant B. The user creates multiple copies of the same BTC and stores it. The user first sends the original BTC to Merchant A and gets the service. Simultaneously, the user sends the copied version of 1 BTC to Merchant B. Since the second transaction was not confirmed by other miners, the merchant accepts the bitcoin and sends the service. But the cryptocurrency that was sent is invalid. This is the case of Double Spending.



Types Of Double Spending Attacks

There are different types of Double Spending attacks:

- **Finney Attack:** Finney Attack is a type of Double spending Attack. In this, a merchant accepts an unauthorized transaction. The original block is eclipsed by the hacker using an eclipse attack. The transaction is performed on an unauthorized one. After that, the real block shows up and again the transaction is done automatically for the real block. Thus the merchant loses money two times.
- **Race attack:** is an attack in which there is a 'race' between two transactions. The attacker sends the same money using different machines to two different merchants. The merchants send their goods but transactions get invalid.
- **51% Attack:** This type of attack is prevalent in small blockchains. Hackers usually take over 51% of the mining power of blockchain and therefore can do anything of their own will.

How Bitcoin Handles Double Spending?

[Bitcoin](#) is one of the most popular blockchains. To combat Double spending it uses some security measures. There are two types of examples of double spending in BTC.

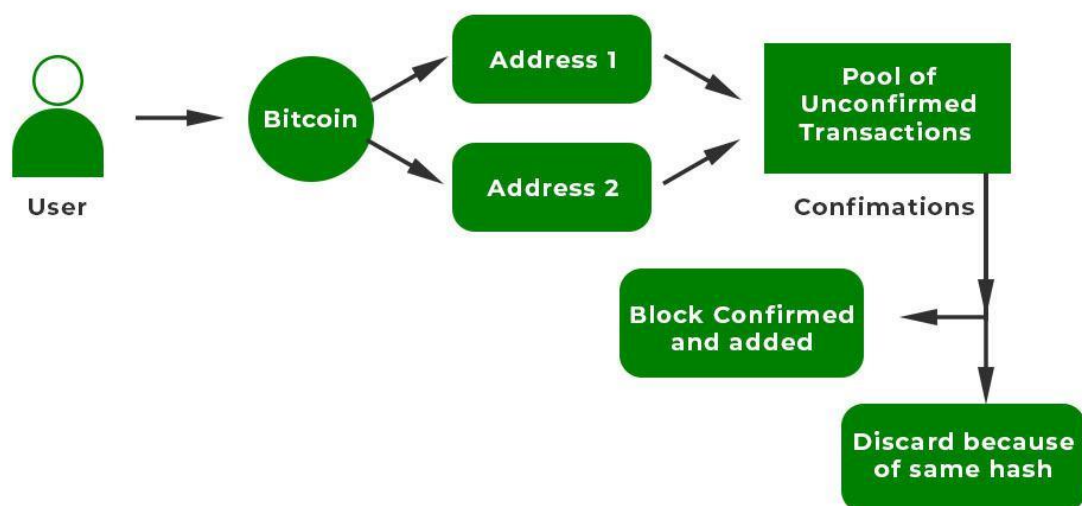
1. The first case is making duplicates of the same bitcoin and sending it to multiple users.
2. The second case is performing the transaction and reversing the already sent transaction after getting the service.

To tackle these double-spending issues, some security measures are taken. They are:

- **Validation:** Validation of transactions by a maximum number of nodes in the network. Once a block is created, it is added to a list of pending transactions. Users send validation for the block. If the verifications are done then only the block is added to the blockchain.
- **Timestamp:** The confirmed transactions are timestamped, therefore they are irreversible. If a transaction is involved with a bitcoin it is verified and done. But in the future, if other transactions are made with the same bitcoin, the transactions will be canceled.
- **Block Confirmations:** Merchants get block confirmations so that they are assured that there was no case of double spending. In bitcoin, a minimum of 6 confirmations are done.
- **Saving copies:** A copy of each transaction is kept at each node so in case of network failure the whole network does not go down.

These security features have reduced double spending to a large extent. Let us discuss a detailed example of how bitcoin handles double-spending.

- A user wants to spend 2 BTC. He/She can create multiple copies of the same cryptocurrency.
- The user can send the same cryptocurrency to two different addresses say 'Bob' and 'Alice'.
- Both of these transactions are sent to the pool of unconfirmed transactions.
- The first transaction T1 would be approved via the confirmation mechanism.
- The confirmation mechanism states that a minimum of six confirmations by miners should be done for block validation. The block is added to the network.
- However, the second transaction T2 didn't get sufficient confirmation so it would be recognized as invalid by the confirmation process. The block with the highest number of confirmations is accepted and the other one is rejected. So transaction T1 is valid, and Alice received the bitcoin.



Blockchain Hash Function

A hash function is a mathematical function that takes an input string of any length and converts it to a fixed-length output string. The fixed-length output is known as the hash value. To be cryptographically secure and useful, a hash function should have the following properties:

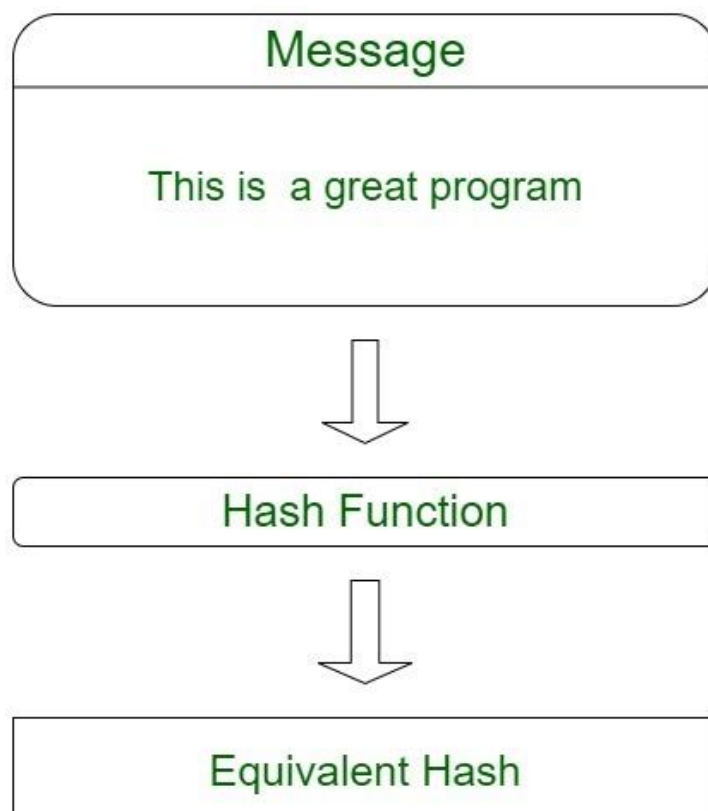
- **Collision resistant:** Give two messages m_1 and m_2 , it is difficult to find a hash value such that $\text{hash}(k, m_1) = \text{hash}(k, m_2)$ where k is the key value.
- **Preimage resistance:** Given a hash value h , it is difficult to find a message m such that $h = \text{hash}(k, m)$.

- **Second preimage resistance:** Given a message m_1 , it is difficult to find another message m_2 such that $\text{hash}(k, m_1) = \text{hash}(k, m_2)$.
- **Large output space:** The only way to find a hash collision is via a brute force search, which requires checking as many inputs as the hash function has possible outputs.
- **Deterministic:** A hash function must be deterministic, which means that for any given input a hash function must always give the same result.
- **Avalanche Effect:** This means for a small change in the input, the output will change significantly.
- **Puzzle Friendliness:** This means even if one gets to know the first 200 bytes, one cannot guess or determine the next 56 bytes.
- **Fixed-length Mapping:** For any input of fixed length, the hash function will always generate the output of the same length.

How do Hash Functions work?

The hash function takes the input of variable lengths and returns outputs of fixed lengths. In cryptographic hash functions, the transactions are taken as inputs and the hash algorithm gives an output of a fixed size.

The below diagram shows how hashes work.



Types of Cryptographic Hash Functions

There are several different classes of hash functions. Some of the popular classes are:

1. RACE Integrity Primitives Evaluation Message Digest (RIPEMD): This set includes RIPEMD, RIPEMD-128, RIPEMD-160, RIPEMD-256, RIPEMD-320.

- Out of these RIPEMD-160 is the most common.
- The original RIPEMD-128 is based on the design principles used in MD4.
- The RIPEMD-256 and 320 have fewer chances of the accidental collision but do not have higher levels of security as compared to RIPEMD-128 and RIPEMD-160.

2. Message-Digest Algorithm: This family comprises hash functions MD2, MD4, MD5, and MD6.

- MD5 is the most widely used cryptographic hash function.
- It is used to generate a 128-bit digest from a 512-bit string broken down into 16 words composed of 32 bits each.
- Ronald Rivest designed this algorithm in 1991 to use for digital signature verification.
- These are no longer considered cryptographically secure methods and should not be used for cryptographic authentication.

3. BLAKE2: It was announced on December 21, 2012. BLAKE2 is a cryptographic hash function based on BLAKE, designed with the aim to replace MD5 and SHA-1 algorithms in applications requiring high performance in software. It provides better security than SHA-2 and is similar to that of SHA-3. It provides the following features:

- It provides immunity to length extensions.
- It removes the addition of constants to message words.
- It simplifies padding and reduces the number of rounds from 16 to 12.

4. BLAKE3: It was announced on January 9, 2020. BLAKE3 is a cryptographic function based on Bao and BLAKE2. It is a few times faster than BLAKE2. This algorithm provides many features like parallelism, XOF, KDF, etc.

5. Whirlpool: It is a cryptographic hash function, first described in 2000. It is a modified version of the Advanced Encryption Standard (AES). Whirlpool produces a hash of 512 bits.

6. Secure Hashing Algorithm: The family of SHA comprises four SHA algorithms: SHA-0, [SHA-1](#), [SHA-2](#), and [SHA-3](#).

- SHA-0 is a 160-bit hash function that was published by the National Institute of Standards and Technology in 1993.
- SHA-1 was designed in 1995 to correct the weaknesses of SHA-0. In 2005, a method was found to uncover collisions in the SHA-1 algorithm due to which long-term employability became doubtful.

- SHA-2 has the following SHA variants, SHA-224, SHA-256, SHA-384, SHA-512, SHA-512/224, and SHA-512/256. It is a stronger hash function and it still follows the design of SHA-1.
- In 2012, the Keccak algorithm was chosen as the new SHA-3 standard.
- SHA-256 is the most famous of all cryptographic hash functions because it's used extensively in blockchain technology. The SHA-256 Hashing algorithm was developed by the National Security Agency (NSA) in 2001.

Uses of Hash Functions in Blockchain

The blockchain has a number of different uses for hash functions. Some of the most common uses of the hash function in blockchain are:

- **Merkle Tree:** This uses hash functions to make sure that it is infeasible to find two Merkle trees with the same root hash. This helps to protect the integrity of the block header by storing the root hash within the block header and thus protecting the integrity of the transactions.
- **Proof of Work Consensus:** This algorithm defines a valid block as the one whose block header has a hash value less than the threshold value.
- **Digital signatures:** Hash functions are the vital part of digital signatures that ensures data integrity and are used for authentication for blockchain transactions.
- **The chain of blocks:** Each block header in a block in the blockchain contains the hash of the previous block header. This ensures that it is not possible to change even a single block in a blockchain without being detected. As modifying one block requires generating new versions of every following block, thus increasing the difficulty.